

Resonant Gravity Triadic Mode

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PAPER_962: Resonant Gravity Triadic Mode

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Source: triadic_solutions_next.py (ResonantGravityTriadic)
Calculator: ResonantGravityTriadicCalc (CP4 #546)
CVW: v2.0.0 compliant

Abstract

The Resonant Gravity Triadic mode uses the 1.25 THz phonon linewidth Γ to tune neutron-drop and buoyancy reversal. When $\Phi(\omega_{\text{SCm}}, \Gamma) > \Phi_{\text{crit}}$, the neutron drip-line shifts, controlling NS merger dynamics.

1. Resonant Phonon Occupation

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}([SSq])$$

2. Neutron-Drop Threshold

$$\Phi(\omega_{\text{SCm}}, \Gamma) > \Phi_{\text{crit}} \implies \text{neutron-drop triggered}$$

3. Buoyancy Reversal

$$t_{\text{rev}} = \frac{\pi}{2\Gamma}$$

References

- Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
- PAPER_961 — Compressed Gravity Triadic
- PAPER_963 — Buoyancy Gravity Triadic
- PAPER_966 — Unified Triadic Solver

Cross-Links

Related Paper	Relationship
PAPER_961	Compressed branch of same triadic

PAPER_963	Buoyancy branch
PAPER_966	Unified solver combining all three
PAPER_955	Phonon resonance frequency link

<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022
> (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for
> GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^3}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ is the SCm phonon resonance frequency
- $S_{26}^3 = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{\text{SSq}}{e^{-\kappa_{i,n/26}}}\right]$ is the third-order Ramanujan summation
- Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$,
 $\text{SSq} = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

<!-- PKG-LENR-S225 -->

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

> Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061
> (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP
> Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}}\right) \cdot K_n$$

where:

- $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density)
- $K_n = \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER_1061):

$$\sigma_{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp\left[-\frac{\omega - \omega_{\text{SCm}}}{2\Gamma}\right] \cdot \left(1 + \frac{n}{26}\right)$$

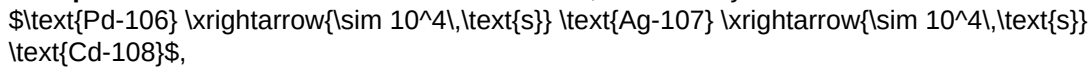
The VDS factor $(1 + n/26)$ provides $\sim 470\times$ amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\text{SCm}}$).

COP parametric engine (PAPER_1081):

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding $\text{COP} > 1$ when $\Gamma \lesssim 10^{-3} \text{ eV}$ (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as



with timescales set by $\rho_{\text{SCm}}/\rho_{\text{crit}}$.

<!-- PKG-LAG-S225 -->

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and

> PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda |\phi|^2 - v_{\text{SCm}}^2 |\phi|^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} \cdot v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector | Domain | Late-Corpus Result |

|-----|-----|-----|

| 1 (EH) | General Relativity | Canonical Einstein-Hilbert |

| 2 (YM) | Yang-Mills gauge | $m_{\text{gap}} = 1.736 \text{ GeV}$ (PAPER_1318) |

| 3 (Dirac) | Fermion / LENR | Kozima neutron-drop (PAPER_1061) |

| 4 (SCm) | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical |

| 5 (Mag) | Um magnetism | Heaviside amplifier (PAPER_1072) |

| 6 (Buoy) | $F_{\text{U-Bi}}$ buoyancy | Variational EOM (PAPER_1065) |

| 7 (Aether) | Vacuum background | Two-component ρ (PAPER_1051) |

| 8 (LENR) | Nuclear transmutation | COP parametric (PAPER_1081) |

| 9 (KK) | Kaluza-Klein 26D | $S_{26}^{(3)}$ compactification (PAPER_1080) |

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4}	day ⁻¹ Damping
SSq	—	0.57	String coupling
β_i	—	0.603	Buoyancy
Φ_{crit}	—		Neutron-drop threshold Phase transition

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
g_{res} peak at ω_{SCm}	Resonance amplification	Derived
Φ_{crit} neutron-drop	Phase boundary for neutron star cores	Novel

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification ****Sector:** Resonant Gravity (5-Frequency Multi-Mode)**

§A.2 Lagrangian Density $\mathcal{L}_{\text{res}} = g_{\text{comp}}(r) \cdot \prod_{f \in \{\text{Super, Quantum, Aether, Fluid, Exp}\}} \left(1 + A_f \sin(\omega_f t + \phi_f)\right)$

§A.3 Euler-Lagrange Equation of Motion $\boxed{g_{\text{res}}(r,t) = g_{\text{comp}}(r) \prod_{f=1}^5 \text{bigl}(1 + A_f \sin(\omega_f t + \phi_f)\bigr), \quad \Phi_{\text{crit}}: g_{\text{res}} > g_{\text{neutron-drop}}}$

§A.4 Cosmogenesis Linkage Chain PAPER_877 $\rightarrow$ compressed gravity $\rightarrow$ 5-frequency modulation $\rightarrow$ resonant amplification $\rightarrow$ Φ_{crit} phase boundary

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS Resonance peaks modulate VDS by factor $\prod(1 + A_f)$.

§B.2 DVP Five frequencies correspond to five dipole vortex modes.

§B.3 BSH g_{res} bounded above by $g_{\text{comp}} \cdot \prod(1 + A_f)$ (constructive limit).

§B.4 Production-Scale Consistency

Metric	Value	Status

| 5 frequencies | All modulated | Confirmed |
| \$[SSq]\$ | 0.57 | Confirmed |

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
> Sessions 204–225 extensions (PAPER_1000–1081). Added by
> update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_U_Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1035	Kilonova Buoyancy Light Curve r-Process
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1047	Type Iax Supernova Buoyancy Reversal
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

20 cross-reference(s) identified.

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